

The empirical recurrence for OEIS A237679, recovered from its tail

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Abstract

OEIS A237679 records “Empirical recurrence of order 67” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 474$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 70$. Its last coefficient is $c_{67} = 148635648 \neq 0$, so no further tail satisfies anything shorter; any order-67 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A237679 is “Number of $(n+1) \times (2+1) \times 0.2$ arrays colored with the upper median plus the lower median minus the minimum of every 2×2 subblock.”. It has offset 1 and begins

474, 4146, 35620, 308814, 2688928, 23395998, 203035400, 1763640726, ...

The entry states:

Empirical recurrence of order 67 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 10:20:50, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 474$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 474$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 67: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 67.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 948$ exact terms from $n = 3$ on, it returns order 67 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{67} c_i a(n-i),$$

c_1	0	c_{18}	16034725999	c_{35}	−8002686942646	c_{52}	−11364030422312
c_2	8	c_{19}	−17999692011	c_{36}	3343745528935	c_{53}	−8174162947252
c_3	227	c_{20}	−47289534632	c_{37}	3862948365146	c_{54}	17820804283880
c_4	2374	c_{21}	53278109472	c_{38}	6699051249929	c_{55}	−6468781561344
c_5	6535	c_{22}	14526885898	c_{39}	−10972413654248	c_{56}	944854003440
c_6	15910	c_{23}	−98695873342	c_{40}	−12456662942988	c_{57}	5311525674208
c_7	−55825	c_{24}	260053432684	c_{41}	2318911130469	c_{58}	−214058595168
c_8	−613891	c_{25}	−144229126739	c_{42}	13546296186212	c_{59}	40975977280
c_9	−1200812	c_{26}	−647543209230	c_{43}	34032223642101	c_{60}	531443827968
c_{10}	−514247	c_{27}	1231842029415	c_{44}	−29749105295496	c_{61}	−116740185344
c_{11}	12565112	c_{28}	297409601529	c_{45}	−23585590243362	c_{62}	−161634574336
c_{12}	60833520	c_{29}	−1924064140093	c_{46}	16388892172070	c_{63}	8911079424
c_{13}	28545096	c_{30}	993987074395	c_{47}	−27695577094683	c_{64}	19418200064
c_{14}	−240345437	c_{31}	−1867351424067	c_{48}	43039369641609	c_{65}	−2798927872
c_{15}	−881409153	c_{32}	−459577498471	c_{49}	8890461802779	c_{66}	−2710831104
c_{16}	−1625434632	c_{33}	8701716035677	c_{50}	−41042190346842	c_{67}	148635648
c_{17}	5157284048	c_{34}	−3570788766023	c_{51}	32328865133480		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 70$, and it is the recurrence OEIS A237679 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{67} - c_1 x^{66} - \dots - c_{67}$. Its constant term is $-c_{67} = -148635648$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the

components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 67 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 67, so $\deg q = 67$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 67, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 17 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 67, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 148635648.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-67 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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